

The Engineer's AI Skills Ladder

The question I hear most from engineers: **what should I be learning?** Here is the ladder I sketch in answer. Five rungs, climbed from where you already stand. Your engineering domain expertise is the ground under it, not a rung, because it is the one part nobody can download.

1**Data engineering**

Get your data clean, structured and where it needs to be. Most automation that fails, fails here first. If you can name where your team's information lives and what state it is in, you are already climbing.

2**Data science**

Make the data answer questions. Not academic modelling: the discipline of asking what the numbers actually support, and being honest when they support nothing.

3**Context engineering**

Feed an AI model what it needs to reason about your problem: the standards, the constraints, the vocabulary of your discipline. This is where domain experts quietly outperform technologists.

4**Harness engineering**

Make it safe to rely on. Evaluations, guardrails, and a human in the loop at the points where judgement and accountability live. The railway will always ask who checked, and someone still signs.

5**Looping engineering**

Systems that improve as they run, where every use makes the next one better. This is where automation stops being a tool and starts compounding.

THREE RULES FOR THE CLIMB**1. Start from your domain expertise.**

It is the asset AI cannot supply. A decade on the railway is worth more at rung 3 than a computer science degree.

2. Find a short feedback loop.

Not a 6 month programming course; motivation fades before it finishes. Pick one task from last week and improve it this week. Then ask: how did last week go?

3. You are not becoming the programmer.

You are becoming the person who knows what good looks like, and can direct the tools that produce it. That judgement is the career.

THE FULLER ANSWER

This ladder comes from a conversation on the TalkCivils podcast: AI for the rail industry, and how civil engineers should think about and use AI. 53 minutes.

Search "TalkCivils Bill Guo" on Apple Podcasts, Spotify or YouTube.

You do not need to reach rung 5 to be valuable. Most teams have nobody standing on rung 1.